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CS313 - DATA MINING AND APPLICATIONS



MEKONG DELTA RAINFALL FORECASTING

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Chapter 1: Project Overview

1.1 Background

The Mekong Delta is Vietnam's key agricultural region, strongly affected by climate change, especially extreme events such as droughts and floods. Accurate rainfall prediction plays an important role in water resource management, agriculture, and disaster warning.

1.2 Data

To model the rainfall process, the study uses Sentinel remote sensing data (reflecting surface and land cover) combined with ERA5 and GLDAS reanalysis meteorological data (describing atmospheric and soil conditions) (1/1/2019 - 31/12/2024) through the Google Earth Engine API. The combination of these multi-layer data sources helps the model capture the relationship between surface – atmospheric – hydrological features, thereby improving rainfall prediction capability.

1.3 Method

Instead of predicting rainfall directly from meteorological – remote sensing data at the current time, the study proposes a two-stage approach:

- Stage 1: Use a deep learning model to predict future meteorological features (e.g., temperature, soil moisture, radiation, vegetation indices,...).
- Stage 2: Use the set of predicted features from step 1 to estimate the corresponding rainfall.

This approach reflects the natural relationship in the climate system, where rainfall forms as a consequence of atmospheric and surface factors in the near future.

Chapter 2: Dataset

2.1 Overview

Sentinel-1

Sentinel-1 is ESA's C-band radar satellite, capturing Earth's surface images day and night, penetrating clouds and operating under all weather conditions. The data are useful for monitoring rainfall, floods, soil moisture, waterlogging, crop growth, and detecting ground deformation. IW mode imagery has a resolution of 10 m, with two polarizations VV and VH.

Sentinel-2

Sentinel-2 is an optical satellite providing multispectral imagery with 13 bands, a resolution of 10–60 m, observing vegetation, soil, water, and urban areas, with data updated every 5 days. The data characteristics include four groups: atmospheric, surface, radiation, and spatio-temporal attributes.

Natural

ERA5 is global climate reanalysis data including temperature, dew point, soil moisture, wind, and surface pressure, directly related to rainfall. GLDAS data supplement physical variables at the land surface such as soil moisture, energy balance, evaporation, temperature, and wind, providing a comprehensive foundation for rainfall modeling.

2.2 EDA



Figure 2.1: Mean of AOT Granule in 2023

AOT increases sharply in April and gradually decreases toward the end of the year, mainly due to post-harvest straw burning and the late-year rainy season (Figure 2.1). The Mekong Delta has only two seasons: the dry season (May–November) and the rainy season, with high rainfall in the late dry-season months (Figure 2.2).

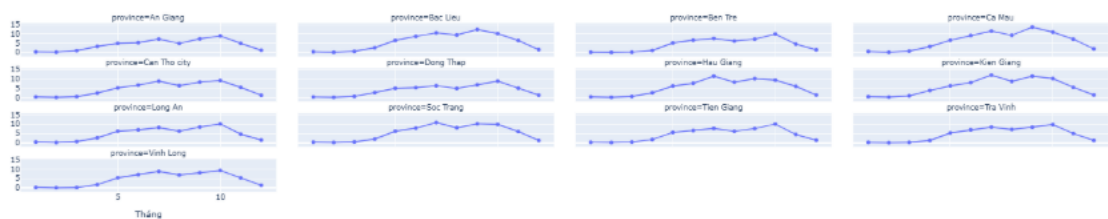


Figure 2.2: Monthly rainfall cycle

Chapter 3: Preprocessing

3.1 Handling null values and dimensionality reduction

Data preprocessing includes removing features with large amounts of missing data (some Sentinel-2 features only appear from 2022) and grouping highly correlated features using PCA to reduce multicollinearity and dimensionality while preserving most of the variance, thereby optimizing model performance.

3.2 Data Sources

The study combines three main data groups: meteorological, remote sensing, and rainfall, all standardized into spatial grid format to support spatio-temporal deep learning. Meteorological variables (temperature, radiation, soil moisture, wind, pressure, etc.) are organized as tensors (C, H, W) , where each pixel represents atmospheric conditions at a location in the Mekong Delta, allowing the model to exploit both spatial relationships and temporal sequences. Sentinel-1/2 remote sensing data are retrieved from the nearest available acquisitions, including NDVI, NDWI, VV/VH, and spectral bands such as RED, NIR..., enabling the model to capture surface and vegetation conditions by stacking each feature as an independent channel.

Rainfall is represented as a single-channel tensor $(1, H, W)$ and serves as the target variable of the entire system. When combined with meteorological and remote sensing data, the complete dataset allows deep learning models to learn the relationship between atmospheric conditions, surface characteristics, and observed rainfall, forming the basis for spatial rainfall prediction.

3.3 Data Splitting

The dataset is divided chronologically:

- 2019–2023: Train
- 01/2024–06/2024: Validation
- 07/2024–12/2024: Test

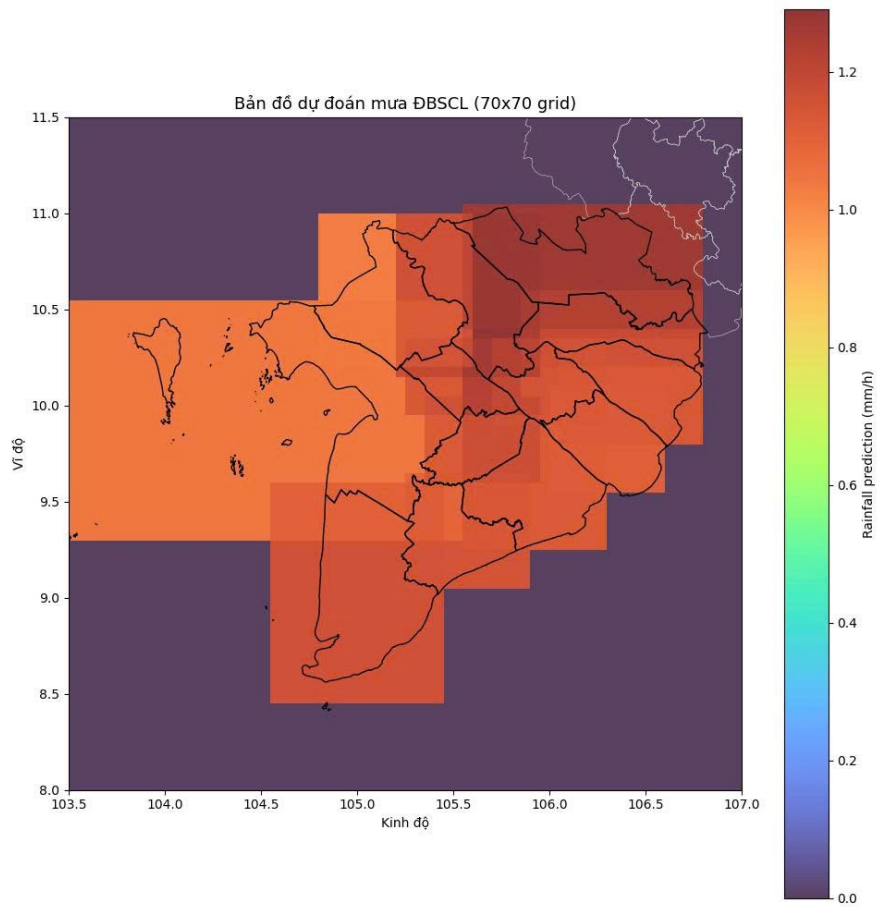


Figure 3.1: Example target

3.4 Data Normalization

Preprocessing applies two steps: (1) Robust Scaling to reduce the influence of extreme values in meteorological and rainfall data; (2) Min-Max Scaling to bring all features into the range $[0,1]$ before feeding them into the deep learning model, helping stabilize training and accelerate convergence.

Chapter 4: Models

4.1 Stage 1

To predict future features, the model must capture spatio-temporal information. The three architectures used include ConvLSTM[1], ConvGRU[2], and TrajGRU[3].

ConvLSTM extends LSTM with convolutional operations in the gating mechanism, enabling the model to simultaneously learn spatial structure and temporal dynamics, suitable for multi-channel precipitation map sequences.

ConvGRU, with two controlling gates and a more compact parameter set, provides faster training and reduces the risk of overfitting in the context of meteorological data with complex distributions.

TrajGRU is a modern model incorporating a dynamic trajectory-learning mechanism, allowing each spatial location to flexibly connect to optimal positions from previous time steps, thereby effectively modeling the displacement and deformation of clouds, rainfall, and non-linear atmospheric structures.

4.2 Stage 2

In the second stage, the future feature tensor from Stage 1 is mapped to generate a 2D rainfall map. The three architectures selected include U-Net[4], Feature Pyramid Network (FPN)[5], and U-Net 3+[6].

U-Net employs an encoder–decoder structure with skip connections to preserve boundary information and local details throughout the compression–decompression process, thereby performing stably in pixel-wise regression tasks.

FPN aggregates information from multiple resolution levels through its feature pyramid structure, helping the model better recognize multi-scale rainfall patterns and increasing sensitivity to localized precipitation regions.

U-Net 3+, a significantly enhanced extension of U-Net, utilizes full-scale skip connections linking all encoder and decoder levels, thus simultaneously leveraging information at all resolutions and improving the reconstruction of rainfall regions with complex morphology. These three architectures enable the assessment of the importance of multi-scale aggregation in spatial rainfall prediction.

Chapter 5: Results

5.1 Overview

Table 5.1 presents the rainfall prediction performance of all model combinations in the two-stage architecture. Stage 1 predicts future meteorological–remote sensing features using spatio-temporal deep learning models (ConvLSTM, ConvGRU, TrajGRU). Stage 2 maps the feature tensor into a 2D rainfall map using three decoder architectures (U-Net, FPN, U-Net 3+). Each model pair is evaluated using two metrics: MAE and RMSE, with bold values indicating the top three highest-performing results.

Table 5.1: Performance of models in the two-stage architecture

Stage 1	Stage 2	MAE	RMSE
ConvLSTM	U-Net	2.84	3.47
ConvLSTM	FPN	2.52	3.64
ConvLSTM	U-Net3+	2.02	2.60
ConvGRU	U-Net	2.92	3.82
ConvGRU	FPN	2.88	4.51
ConvGRU	U-Net3+	2.33	2.92
TrajGRU	U-Net	1.69	2.05
TrajGRU	FPN	2.66	3.63
TrajGRU	U-Net3+	1.59	2.06

5.2 Stage 1

TrajGRU is the highest-performing model in the entire architecture. With the lowest MAE and RMSE across the Stage 2 architectures, TrajGRU demonstrates that its learnable trajectory mechanism is particularly well suited for modeling the spatial–temporal evolution of rainfall.

ConvLSTM ranks second in overall effectiveness. Although it captures temporal dependencies well, its fixed convolutional structure limits its ability to represent nonlinear motion and complex rainfall systems.

ConvGRU yields the lowest performance. Its smaller parameter count makes the model lightweight and fast to train, but this also restricts its capacity to capture deeper spatial–temporal interactions, especially in multi-channel meteorological sequences.

5.3 Stage 2

U-Net 3+ outperforms the remaining architectures in most combinations. With its full-scale skip connections, U-Net 3+ aggregates features across all resolutions and reconstructs rainfall intensity at the pixel level with high fidelity.

U-Net provides moderate and stable performance. While it preserves spatial features effectively, the traditional U-Net remains limited in multi-scale propagation, particularly when the data exhibits complex rainfall structures.

FPN does not perform well in the context of rainfall map regression. Although FPN excels in object detection, its feature pyramid design prioritizes contour and structural recognition, making it less optimal for continuous map regression.

5.4 Conclusion

Experimental results show that TrajGRU is the strongest Stage 1 model due to its ability to effectively model spatial–temporal motion, while U-Net 3+ consistently provides the highest accuracy in Stage 2 for reconstructing rainfall maps. The combination of TrajGRU and U-Net 3+ achieves the best overall performance.

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